
Teaching statistics to doctoral students with Lonergan's insight-based critical realism

Charles T. Tackney* and Wencke Gwozdz

Department of Intercultural Communication and Management,
Copenhagen Business School,
Porcelænshaven 18A,
Frederiksberg, 2000, Denmark
Fax: +45-3815-3840
E-mail: ct.ikl@cbs.dk
E-mail: wg.ikl@cbs.dk
*Corresponding author

Abstract: The authors report on the development, implementation, and evaluation of a doctoral level intensive quantitative methods course featuring an integrated orientation to, and practicum in, statistics competence for qualitative researchers. A feature of this integrated approach is the inclusive point of course departure: the general empirical method offered by the insight-based critical realism of Bernard J.F. Lonergan. This epistemology grounds curriculum design, wherein key statistical concepts are presented in the very first session and reinforced throughout the intensive. The practicum component offers guided study in the statistical use of SPSS using a common EU data set. Course evaluations indicate students who had previously felt disinterested or unaware of the significance and role of quantitative studies emerged from the three day intensive with a better understanding and sense of empowerment in both the basic use of statistics as well as the complementary nature of quantitative and qualitative research methods.

Keywords: doctoral curriculum; statistics instruction; research methods; insight-based critical realism; Bernard J.F. Lonergan; epistemology.

Reference to this paper should be made as follows: Tackney, C.T. and Gwozdz, W. (2014) 'Teaching statistics to doctoral students with Lonergan's insight-based critical realism', *Int. J. Management in Education*, Vol. 8, No. 1, pp.1–21.

Biographical notes: Charles T. Tackney is an Associate Professor in the Department of Intercultural Communication and Management of Copenhagen Business School. He was the Programme Director and Chair of the CBS Asian Studies Programme (2001–2006). His research interests include Japanese and comparative industrial relations, theology of the workplace issues, and research methods implementation of insight-based critical realism based on the works of Bernard J.F. Lonergan. Along with department colleague Maribel Blasco, he co-founded the 2012 Researchers in Management Education (RIME) Research Cluster: <http://www.cbs.dk/RIME>.

Wencke Gwozdz is an Assistant Professor at the Department of Intercultural Communication and Management (ICM) at Copenhagen Business School. She holds a Doctorate in Economics and Social Sciences from the University of Hohenheim, Stuttgart, Germany. Her main research focus is on sustainable development, in general, and sustainable consumption, in particular. Beyond

economic and psychological micro models explaining consumer behaviour, she is interested in consumers' role in business, society, and governance as well as in policy implications. She teaches mainly in quantitative methods and consumer behaviour.

1 Introduction

In the spring 2010 term, the authors successfully ran a three-day intensive research methods course for doctoral students at Copenhagen Business School (CBS). Open to students throughout Europe, the course received 26 registered students – all of whom attended and completed the course. The course was designed to fill a particular niche: help qualitatively-oriented doctoral students enhance their skills through a practicum given an underlying curriculum design that welcomed a diversity of qualitative approaches and variance in prior statistics study.

This is a practitioner's curriculum report of initial development, deployment, and course evaluative outcome. Course success derived from three features, detailed below. First, the insight-based methods approach of Bernard J.F. Lonergan enabled an orientation to general methods that is welcoming of diverse qualitative perspectives. Second, Lonergan's epistemology grounded the insight-based intensive design, wherein key statistical concepts were presented first, with reinforcement occurring throughout the subsequent SPSS practicum. Third, common student exercise use of a current EU data set from descriptive to inferential statistics offered ample time for collaborative student teamwork and discussion. The practicum success was enabled by a clear set of instructional slides on each statistics step and an instructor very familiar with both the EU data set and the instructional slides.

1.1 Background: curriculum need and theoretical design

On 15 April 2009, a lunch meeting was held at our Department of the Copenhagen Business School (CBS).¹ PhD school consortium staff spoke about a quantitative methods course need. The perceived need was specific; the course would be brief, intensive, and offer practical numeracy skills for doctoral students with little prior background in statistics.

The co-authors were then involved in research methods instruction at both undergraduate and Master's levels in CBS, collaborating in course design and instruction. With the proposal deadline in late August, we began planning by learning more about expectations, current offerings, and the quantitative skill level of current doctoral students.

The PhD school administration had an idea about a niche course that would run in the manner of an intensive lasting between two and five days. This was due to standard practice, reflecting an appreciation for the budget and travel costs incurred by students attending from a distance. Two key issues emerged. First, the intensive course needed to start from a very basic level, while offering 'hands-on' statistical skills training and hypothesis-testing experience. There were several statistics courses already on offer; these tended to be rigorous. Some were advanced, obliging considerable prior

quantitative coursework. Others were designed to take up an entire semester, such as the courses found in a standard economics or mathematics department offering.

Thus, the envisioned course design would have to be able to presume, as an initial point of departure, student participants with no prior statistics or returning students with considerable time between earlier studies and present doctoral research concerns. In addition, the skilling goals of the course were expected to bring students to a point of being able to conduct and interpret independent software-based data analysis – or at least to better recognise salient questions in such interpretation. The second issue was related to the first: the envisioned student group was expected to be professionally very diverse and theoretically sophisticated. Thus, on what basis would such methods diversity find common ground that aimed at initiating and/or refining quantitative skills for predominately qualitative doctoral students?

From a knowledge management perspective, the course was in search of a common theoretical platform from which the statistical studies could be successfully launched; this is the shared site of learning Nonaka refers to as 'Ba', from the Japanese word for 'place' (Nonaka and Takeuchi, 1995; Nonaka, 2005). While Nonaka's concept derived from shop floor learning practices in Japanese manufacturing, this course needed a functional equivalent at the intellectual level of graduate instruction. We had to design and present an inviting epistemology sufficiently robust to be informed by, while also informing, any given social science orientation that might present itself in the potential student audience of contemporary European scholarship.

As educators involved in tertiary curriculum design, particularly for research methods instruction, we sought to document course development and delivery as the process unfolded. Our goal was to craft a course that would itself manifest and be subject to the norms of the epistemological and practicum teaching 'place' envisioned. This would provide a degree of pedagogic reflection and reflexivity capable of including, and being reported to, student participants themselves from the first year and thereafter (Dee and Daly, 2009). Over time, should the course effort succeed, it would itself become a matter of research interest suitable for reporting in conference venue, with a view to disseminating insights gained and receiving broader feedback. Against this summary of two design issues to be faced and curriculum research methods motivational goals of the co-authors, let us look at the method that informs this paper.

2 Method

This is a practitioner's report about a presumed educational market need in Northern Europe, the particular curriculum design issues and implementation constituting our course response, and reflections on future prospects. The educational market niche concerned graduate qualitative student orientation to and skilling in general empirical method with practical quantitative instruction and training in computer-based statistical analysis. The co-authors were responsible for course design and instruction. As a practitioner's report, we intend to provide sufficient detail that this course may be replicated in, or adapted to, other settings. Thus, the paper becomes a benchmark document for curriculum design history. To achieve this, our data section:

- a briefly summarises the chronology of curriculum development
- b addresses curriculum design concerning two issues of critical import:

- the epistemological platform or point of departure
 - the approach to the statistics practicum of instructional training.
- c reports the pre-course survey data findings
- d details course enactment
- e records the course evaluation data.

The discussion section notes outcomes of the course, additional reflections and information concerning key issues of curriculum design, and specifies revision points for subsequent course offerings.

3 Data

3.1 *Chronology of course development, implementation, and evaluation*

An initial course proposal was submitted to the Doctoral School staff on 28 August. Approval, with suggestions, was obtained by 3 September. The course was within CBS and internationally. The posted course schedule and learning objectives are given in Appendix 1. At the registration deadline of 12 April, 26 students had registered and the instructors were informed that the course would proceed. The number of registered students matched the largest PC-equipped classroom in the institution and was the highest registration for a doctoral course in the programme's history.

As part of the course design, the instructors developed a survey for registered students to ascertain their quantitative background and course expectations. The survey was created 19 April on the Survey Monkey website.² There were 19 completed responses and 23 responses in total by the time the course commenced, all done between 20 April 20 and May 7, 2010. Data from this survey is summarised below in data section D.

We next take up the two issues considered important for course success. First, we explain how the course would be oriented to students from different academic backgrounds and with different levels of statistics experience. Then, we consider the statistics practicum.

3.2 *Insight-based critical realism and general empirical method*

Our first curriculum design challenge concerned the basic course orientation. Ours was to be a diverse group of qualitatively oriented students with little or no prior exposure to statistics. How might we instantly construct an intentional community?

More than most, social scientists benefit from a thorough grounding in epistemology. Graduate students learn epistemology either directly, through coursework in philosophy, or indirectly, as an implicit component of specialised coursework in a particular academic trajectory. Yet, from economics to specialised discourse analysis found in socio-linguistics, all proceed from epistemological presumptions: the various and varied premises that serve foundational purposes.

While graduate study involves mastery of one's field of specialisation in the social sciences, graduate students inevitably encounter challenges to their work from scholars with different training, who sometimes possess profoundly different epistemological

orientations. Thus, thorough grounding in a theory of knowledge particularly appropriate to the social sciences, offering a general empirical method, would seem an appropriate point of departure for qualitatively oriented students seeking greater facility in quantitative studies.

This general empirical method is a feature of the insight-based critical realism developed by Bernard J.F. Lonergan (1971, 1988, 1992, 1996a, 1996b). For the social scientist of any particular specialisation – indeed, anyone who is attentive, intelligent, reasonable, and responsible – Lonergan's epistemology begins with the simple invitation to reflect on the nature of insight. What scientist does not think about this elusive, key feature that lies at the very heart of discovery?

Lonergan's epistemology is a form of critical realism. However, it significantly differs from the critical realism developed by Bhaskar and others, a stream more familiar to management scholars (Collier, 1994; Bhaskar, 2008; Archer et al., 1998). Bhaskar's approach towards transcendental realism, "regards the objects of knowledge as the structures and mechanisms that generate phenomena; and the knowledge as produced in the social activity of science" (Bhaskar, op. cit., p.24). As useful as this approach may be, it cannot successfully offer a common grounding for a diverse graduate student group on two counts. First, the 'regard' taken towards objects and mechanisms does not yet actually *know them to be so*. Thus, a mixed student group with varied academic 'priors' may or may not find this option of 'regard' in respect to quantitative methods as in any way compelling. Second, and arguably more important, the 'regard' Bhaskar wrote of does not adequately explain its own presumptive validation.

In sharp contrast, Lonergan begins with the inescapable data of consciousness.³ The human subject is invited to be attentive, alert, reflective, and able to judge issues, particularly and precisely for her/himself. From this invitation, a process of self-appropriation may proceed. This process centres on Lonergan's (1988, 1992) exploration of insight as an aspect of cognitional structure. Reflection on insight reveals five characteristics. First, it comes as a release from the tension of inquiry. Second, its arrival is often very sudden and unexpected. Third, it occurs as a result of inner conditions, not outer circumstances (although these may either aid or hinder insight). Fourth, insight pivots between abstract issues and concrete circumstances. And, finally, insight, once obtained, "passes into the habitual texture of the mind" [Lonergan, (1992), p.28].

From this reflection on a key element of general understanding, the patterns of data analysis, concept formation, reflection, and classification readily proceed from a single, simple premise: humans evidence an unrestricted desire to know, to understand. Lonergan refers to this reality as 'the fact of inquiry' and characterises the point dramatically; "Deep within us all, emergent when the noise of other appetites is stilled, there is a drive to know, to understand, to see why, to discover the reason, to find the cause, to explain why" (Ibid., p.28).

Note that this understanding speaks less of any specific datum in consciousness than the cognitional structure *of* consciousness. Consistent with our unrestricted desire to know, reflection reveals various structured cognitive anticipations of understanding. Lonergan makes an important distinction between classical and statistical heuristic (anticipatory explanential) structures. The former seeks the function and systematisation of verifiable possibilities, which we nominally refer to as laws. The latter, in contrast, seeks to:

- identify verifiable possibilities
- explore correlations
- approximate concrete processes
- grasp probabilities
- gain predictable control over these probabilistic outcomes (Lonergan, 1992).

These heuristic structures are complementary: observed statistical patterns of significance may confirm causal association or provide reasonable grounds to reject the validity and reliability of long-presumed causal relationships. Lonergan summarises this point, “Now the relations between data must be either systematic or nonsystematic. It follows that in any given case either the classical or the statistical anticipation must be correct” (Ibid., p.128).

As the initial presentation of these cognitional structures proceeds, informal review of basic research methods terms can be introduced as needed. This offers opportunity to remediate gaps in individual knowledge in a discursive, non-threatening manner, bringing participants to a shared sensibility and common language of research method.

Two additional points help establish the common epistemological platform. First, we need to clarify what a ‘statistical pattern of significance’ actually means. This can be done by delving into two notions: the sampling distribution of the mean and the central limit theorem. The former refers to the plotted frequency distribution of means from many samples of a statistic. The latter builds upon the former concept by noting how repeated plotting of frequency distributions of these sample statistics will approximate a normal distribution of data as the same size increases.

To sharply focus a diverse student group on this common task, a *Washington Post* newspaper article about the emerging patterns of SARS deaths was sent to each participant one week before the first class, along with a set of questions (Vedantam and Stein, April 25, 2003, see Appendix 1). The text takes up issues of death rate variance, estimation errors and the relation between sample size and sampling validity. The course would begin with class discussion of text and questions. Through this concrete exercise, key statistical issues are identified, the insight goals become known, and the instructor would quickly learn the statistical competence of each participant.

Through presentation of this introductory material on insight, offered in a lecture/discussion following the SARS article discussion, the intensive course foundation would be established; the class would concern itself, as an intentional community, with the nature of data and statistics. As Morelli wrote, “Whether or not one is to become an empiricist, a pragmatist, an idealist, and so on is not something to be settled *a priori*; rather, it emerges, in Lonergan’s view, as a reflective, philosophical conclusion from the thoroughgoing analysis of the spontaneity of human consciousness” (1999, p.177). Students would next begin direct use of SPSS for data analysis, with a view to enacting – for personal insight – exercises that will ‘fill in’ their grasp of the principles governing statistical significance.

This curriculum design serves explicit as well as tacit strategic functions. The explicit dimension concerns diverse student focus upon insight and what this specifically means for statistical principles: the grasp of a normal distribution and the central limit theorem. The tacit dimension involves the student’s focused desire to know and understand as this motivation personally commits them to the intensive course.

Pedagogically, this focused course beginning recognises the ubiquity of statistical computing programmes. It is simply no longer necessary for students of a qualitatively applied disposition to have to learn rote statistics in the inductive manner of the past. Intellectual insight into these key statistical principles at the very outset, when promptly applied in these programmes, appears to be an effective skilling approach.

3.2.1 Statistics practicum: design of a statistics intensive with guided exercises

The statistics practicum was designed to build on whatever knowledge students had acquired, relying on their participation and mutual help. The pre-course survey helped to closely tailor initial course plans to student needs – and this even before the course commenced.

We planned to first cover basic statistical knowledge: overview of quantitative methods, data types, sampling, and descriptive statistics. The course description and lesson plan are given in Appendix 2. Students would then be introduced to a common data set. The data set is the German Socio-Economic Panel (GSOEP), from the German Institute for Economic Research DIW (Haisken-DeNew and Frick, 2005). We prepared a subset of the 2002 GSOEP.

Following rapid descriptive statistics treatment, where a theory presentation is followed by student data set enactment, we intended rapid steps to inferential statistics, up to and including regression and factor analysis. That said, each step permits repetition and reinforcement of basic terms, from different perspectives. Data and method requirements would then be highlighted, including the various assumptions different statistical measures oblige. Participants would thus learn the rationale behind statistical methods without undue focus on formulas. We thought it more important to aid understanding of what the various methods offered than presenting mathematical proofs.

The practical part of the statistics sessions were designed as follows. An initial session of review and remediation consisted of two parts:

- a basic statistics knowledge: making sense of available methods, primary versus secondary data, sampling issues as well as other important statistical terminology
- b an introduction to SPSS.

While the first part ensured a minimum level of required knowledge of each student, the second part got across the structure of the statistical software to be used. At the end of this session, students would work on proficiency by immediately creating new variables and entering data.

In the second session, we planned to deal with descriptive statistics that could be immediately developed with SPSS. The descriptive statistics were frequencies, measures of central tendency and measures of dispersion. The hands on exercise had several intentions, among them to:

- apply the descriptive and read and interpret the output
- showing ways of presenting data in graphical and tabular form
- creating or recoding new variables, as well as
- filtering sub samples.

Student performance in handling of data to run descriptive statistics should be extremely successful, as this is not only for its own purpose, but necessary for higher order analysis. Yet, these steps are quickly learned and the learning readily apparent. For example, participants would be asked to recode and then compute variables to get descriptive statistics for female and male net income.

In the third session, we hoped to take up differences between descriptive and inferential statistics, and consider the main purposes of the latter. This would occasion another look at sampling and sampling issues. As a first method, we would introduce the t-test for testing differences. Taking up the example from the descriptive statistics session on income differences between the sexes, participants would gain insight into why inferential statistics are meaningful and that it makes a difference whether one speaks of a sample or a population.

Practicum sessions would continually, overtly link back to the opening SARS discussion and presentations, with each exercise tailored to sequentially advance statistical topics of interest. The session on t-tests serves as an example. The theoretical statistics component would offer an overview of various methods for testing differences, the rationale of t-tests for two independent samples and an example t-test – including output tables from SPSS and their interpretation. In the hands-on session, students generate descriptive statistics on individual monthly net income. Next, each student produces descriptive statistics for monthly net income for men and women and then compares the average income of both sexes. Each student works individually, but neighbourly help is encouraged. Both instructors remained available throughout.

Discussion would, for example, take up why a t-test is needed if one already notices a difference in the sample; this reverts to the meaning of inferential statistics that had been discussed in the first day's lecture. A t-test on the difference of monthly net income of men and women would be conducted and the output tables discussed. To improve data handling, a final task in these exercises explores income differences between high and low educated women, where students filter the data by sex before the t-test. Throughout, discussion would arise as needed, on the presumption that one question was likely a proxy for others' concerns. To aid homework, a step by step guide, created by one of the co-authors, was provided with the exercises. The guide features screenshots of SPSS.⁴

The fourth session would focus on testing relationships, particularly correlation and ordinary least squares (OLS) regression. These exercises, too, included repetition of previously learned content. Correlation and regression SPSS exercises output would be interpreted and discussed in detail. Additionally, ways of presenting the results in publications could be taken up.

In the final session, we recruited a guest lecturer – a colleague from the department – to speak about his experiences in empirical research, especially in respect to his use and struggles with factor analyses. He would speak of the path from a research idea to conference paper and journal publication. We hoped such a presentation and discussion, against the background of intensive theory and statistics practice, would offer concluding insights for doctoral students at a critical juncture in their academic pursuits.

This summarises our initial course design. We thought the pace would be mainly determined by students' needs. The goal was to create in-depth understanding for the basic logic of statistics, not simply rush through a lesson plan.

3.3 Student registration and pre-course survey data

We invited the registered students to complete a survey before the course began (Appendix 3) as an aid to comprehend the target group and their needs. As evident from the questions, we thought a successful intensive course required us to incorporate:

- a existing statistical knowledge
- b how far quantitative methods could be pursued
- c what the participants thought they wished to learn.

Eighteen out of 26 registered students completed the questionnaire. The group was very heterogeneous regarding scientific background: participants' Master's degree varied from classical business education to art/design, IT, or food science. Some 44% of the participants had no semester of statistics in their academic career, 44% had one semester of statistics and 12% had three or more semesters. Only a third indicated an understanding of the difference between descriptive and inferential statistics.

The second part of the questionnaire was about use of quantitative methods in students' research. Eleven respondents intended to use quantitative methods to complement their qualitative research. Seven respondents only wanted to participate in the course for a better understanding of quantitative research, without any particular intent to use it. Most of the eleven respondents intending to apply quantitative methods wished to use an existing survey. Among these, most did not know how to analyse their data in respect to variable and methods specification. About seven students had already established a data set of interest.

The third part of the survey concerned what the students knew and what they wished to learn about statistics. Three did not know the difference between primary and secondary data. Moreover, we asked about the dependent variable – that is, their research variable of interest. There were 11 students able to specify this. To get a more concrete idea about their intended measurement, we asked about its scaling. About 50% of the participants stated that they understood the point of the question and this was an issue of course interest.

Overall, we had a very heterogeneous group with many participants being uncertain about how to understand and use quantitative methods. More than half of the course wanted help in applying methods within their research project. The survey results supported our initial sense that we would have to begin the course from the very basics of statistical knowledge.

3.4 Course enactment

The course began in May 2010 and ended as scheduled. There were 26 registered students, all of whom arrived, attended, and remained. While the course was intended as a doctoral course, a number of Master's students also signed up and participated. Considerable time was devoted to ensuring that non-CBS students gained access to the computers in the classroom; each visiting student required three separate passwords for entry to the PC station, the SPSS software, and the intranet website. While computer room carrying capacity was stated to be sufficient for an enrolment of 26 students, equipment failures required some students to double up or resort to using their own laptop or netbook, with this obliging yet another CBS system access password.

The two instructors had distinguished who would be responsible for what session; team teaching had not been envisioned. In fact, partly due to the teamwork required to overcome the CBS IT logistical difficulties, we were together for almost every session. The designated instructor would assume the position at the lectern or PC, while the other went to the back of the room, listening and taking notes, offering comment when helpful, and otherwise multitasking.

At the beginning of the course, we circulated a sign-up sheet and invited those who wished to briefly present this research agenda and data to volunteer. Three students expressed interest and were scheduled. One student who had not signed later expressed an interest in doing so. In her presentation, she reported that the course had inspired her to see a quantitative analysis slant in her data set she had not previously observed. On presentation, it became apparent that she had a unique set of data, and that it was the entire population, no mere sample. This resulted in a useful discussion about the comparative merits of sample versus population data sets.

The instructors discussed progress after each session, reviewing course goals with actual student performance, interest, and pacing. Due to the time taken up in more basic issues of statistics, the practicum planned for factor analysis was dropped. The statistics instructor offered a brief 'bridge' about the nature and role of factor analysis by way of introducing our final session's guest speaker. His presentation took up from this transitional opening because he began by noting his own introduction to factor analysis arose from a particular data set he was struggling with – the very data set that was his presentation theme. This guest presentation also offered useful, concrete reflection on the entire research cycle: from initial notion to academic journal publication. The reflection was sufficiently concrete that the presenter shared freely of insights gained and obstacles encountered, resulting in a useful class discussion that summarised the entire course experience.

3.5 *Course evaluation data*

Twenty four students handed in a course evaluation of the standard type used in the CBS; the two absentees had to leave early due to travel obligations. The overall evaluation was very positive: 22 of 24 students evaluated the overall quality of the course as good or excellent. Other aspects such as the practical organisation and the course design got high ratings. Only the choice of literature suffered, in that there were but nine students who expressed satisfaction with what was provided. Others indicated this aspect as a course weakness. Our intention, on completion of one successful effort, was to compose and provide a compendium, if the course was to recur. One student was not satisfied with any aspect of the course, recording the second lowest possible category on each and every measure. This was surprising, as nothing was ever stated or apparent throughout the intensive.

Student comments on weaknesses of the course suggested it should be longer, with more hands-on sessions. They also suggested more time be given so that students could simply 'play around' with SPSS on their own.

On the positive side, a student stated: "A lot of useful stuff in a very short time (no need for years of math)". The hands-on session was often positively mentioned in the context of the whole course design. One student wrote, "A strong aspect of the course was the practical part, whereby we had the opportunity to apply some theoretical concepts using a particular software. Furthermore, I think (one instructor) was very good

in explaining what in fact those theoretical concepts represent". In addition, the pre-survey to gather student knowledge and needs was appreciated by students. One wrote, "Course is adapted to student needs and context. Good with pre-course survey...." We later learned that the course evaluations were the most positive set of results from any doctoral course offered by the particular doctoral school unit.

4 Discussion

If we consider this doctoral level quantitative course offering in light of initial plans, the outcome was very successful – that one negative evaluation notwithstanding. Registration far exceeded expectations. Attendance and evaluation statistics suggest that those who participated found the course useful, despite and possibly due to their very diverse academic profiles and stated interests. The course was initially offered on a provisional basis, with no certainty of market need and little expectation of re-offering the course. Yet, in a follow-up meeting with the Doctoral School Director on 4 June, the decision was made to regularly offer the course as one of seven 'mandatory course electives'. It continues to be offered, most recently in April 2012.

One important issue for reflection concerns the praxis achieved in the very first lecture between the role of insight, which served as the point of departure for Lonergan's critical realism, and the early presentation of two critically important statistics concepts: the sampling distribution of the mean and the central limit theorem. Grasping the notion of insight itself, as Lonergan variously notes, is a self-appropriation of personal understanding. From course instruction at undergraduate and Master's levels, this notion of personal insight appears very helpful for students. For many, it seems helpful to introduce this early on in particular courses, with students then becoming more aware of their personal ownership and possession of insights gained (Tackney et al., 2010).

At the doctoral level the instructionally proximate link between the nature of insight and the introduction of two challenging statistical principles appears to have been particularly self-reinforcing for the class participants. In other words, it was possible to present statistics in a manner that even those who delightedly 'hate' quantitative studies can find understandable and, within the same day, intuitively comprehensible. This sets in motion thought to practical uses of these insights gained.

Furthermore, insofar as the epistemological 'platform' upon which insight builds is based on cognitional structure, the entire nature and manner of thinking about being a reflective and reflexive practitioner is subject to a helpful modification. Qualitative researchers, from whatever particular skills set they may have acquired and arrive to the course possessing, appear to abruptly 'lose' the sharp factional allegiances that were previously informing their intellectual horizon. This frequently comes up in regards to whether or not a student 'likes' quantitative or qualitative studies.

Prior generations of statistics instruction – before the now commonplace of personal computers and statistics software – obliged delving into the mathematics of statistics *because each and every student had to individually learn and know the mathematical details in order to enact the statistical 'moves' for their own research praxis*. Obviously, this is no longer the case: Instead, this course suggests we need only provide the theoretical data in an intelligent manner so that the statistical software tools can be correctly deployed, the results properly appreciated. Thus, appropriate preparation and

provision of data and its post-analysis interpretation are the appropriate, and only, contemporary researcher tasks to master.

As an ironic result, it should become possible to present quantitative analysis to qualitative researchers in a manner that simply, clearly, directly and immediately appeals to their qualitatively-oriented academic curiosity. Put simply, statistics has become a thinkers' game – even and especially for qualitative researchers. This suggests an intensive course such as ours may be usefully offered even to colleagues, not only students, who may not have studied statistics earlier and wish an efficient, appropriate intensive experience.

In the course design, we appear to have shown that the two principles that ground inferential statistics – the sampling distribution of the mean and the central limit theorem – can readily be presented to students with little prior knowledge or interest in the mathematical complexities of the subject. The principles need only be presented in proximate reference to the role of insight in understanding. This creates a particular and fascinating circumstance wherein the role of statistics – as a research tool – becomes apparent as a useful and relevant matter for intellectual inquiry for qualitative researchers.

Given these reflections, it would appear that the opening orientation – this 'insight-statistics principles' nexus – is particularly useful, because it is initially challenging. As the course evaluations suggest, students evidenced a deepening understanding of both as the intensive course proceeded, even to the point of one student presenting on a quantitative analysis data stream that had previously not been recognised as offering such a path for analysis.

Institutionally, what had been first offered on a very tentative, exploratory basis has become a dedicated, annual course offering as one of seven 'mandatory electives.' Practically, we first offered the course to the maximum carrying capacity the CBS could manage: about 30 students, for a standard computer room. This proved to be too large, both in view of IT challenges as well as encouraging student participation. In the future, the class size will not exceed 15 or 16 students. We will need to streamline the password acquisition process for guest students, a matter the CBS should see as worthwhile.

Student feedback indicated that some material was slow to be provided, or simply absent. In part, this was due to instructor surprise that the course would run. But, having run once, materials can be more readily provided. In student written and spoken comment, it became clear that more background information on Lonergan's critical realism would be useful. To that end, a brief annotated bibliography has been created (see Appendix 4).

That the intensive proceeded as a team-taught course throughout was not initially intended. The CBS has an instructional tradition of designated course coordinators, who then craft courses that call upon other colleagues to participate. Course descriptions and lesson plans are then approved by Study Boards composed of faculty and elected student representatives. The coordinator is not provided professional time to observe each lecture given and rarely, if ever, does so.

Thus, locally, it would be unusual for one instructor to teach an entire course by her/himself. In principle, however, the course could be readily replicated by anyone with a modest grasp of philosophy and statistics. Furthermore, the critical nexus of insight, presented in close proximity to the two principles of inferential statistics, can be usefully offered even at the undergraduate studies level (Tackney et al., 2010). Despite these points, the repartee which obtained from the presence of both instructors throughout the

course offered a depth and richness the students evidently enjoyed. The course was an adventure, and this encouraged continuous participation by its co-creators.

References

- Archer, M., Bhaskar, R., Collier, A., Lawson, T. and Norrie, A. (1998) *Critical Realism: Essential Readings*, Routledge, London.
- Bhaskar, R. (2008) *A Realist Theory of Science*, Routledge, London.
- Collier, A. (1994) *Critical Realism: An Introduction to Roy Bhaskar's Philosophy*, Verso, London.
- Dee, J.R. and Daly, C.J. (2009) 'Innovative models for organizing faculty development programs: pedagogical reflexivity, student learning empathy, and faculty agency', *Human Architecture: Journal of the Sociology of Self-Knowledge*, Vol. 7, No. 1, pp.1–23.
- Haisken-DeNew, J-P. and Frick, J.R. (2005) 'DTC – desktop companion to the German socio-economic panel (SOEP)', Discussion Paper, DIW, Berlin.
- Lonergan, B.J.F. S.J. (1971) *Method in Theology*, 2nd ed., University of Toronto Press, Toronto.
- Lonergan, B.J.F. S.J. (1988) 'Cognitive structure', in Crowe, F.E. S.J. and Doran, R.M. S.J. (Eds.): *Collection*, Vol. 4, pp.205–211, in the Collected Works of Bernard Lonergan, University of Toronto Press, Toronto.
- Lonergan, B.J.F. S.J. (1992) *Insight: A Study of Human Understanding*, 5th ed., edited by Crowe, F.E. S.J. and Doran, R.M. S.J., Vol. 3 in the Collected Works of Bernard Lonergan, University of Toronto Press, Toronto.
- Lonergan, B.J.F. S.J. (1996a) 'Insight revisited', in Ryan, W.F. and Tyrrell, B.J. (Eds.): *A Second Collection: Papers by Bernard J.F. Lonergan*, pp.263–278, University of Toronto, Toronto.
- Lonergan, B.J.F. S.J. (1996b) 'Theology in its new context', in Ryan, W.F. and Tyrrell, B.J. (Eds.): *A Second Collection: Papers by Bernard J.F. Lonergan, S.J.*, pp.55–68, University of Toronto, Toronto.
- Morelli, M.D. (1999) 'Lonergan's unified theory of consciousness', *Method*, Vol. 17, No. 2, pp.171–188.
- Morelli, M.D. (2003) 'The realist response to idealism in England and Lonergan's critical realism', *Method: Journal of Lonergan Studies*, Spring, Vol. 21, No. 1, pp.1–23.
- Morelli, M.D. and Morelli, E.A. (1997) *The Lonergan Reader*, University of Toronto Press, Toronto.
- Nonaka, I. (2005) 'Managing organizational knowledge: theoretical and methodological foundations', in Smith, K.G. and Hitt, M.A. (Eds.): *Great Minds in Management: The Process of Theory and Development*, Chapter 18, pp.373–393, Oxford University Press, Oxford.
- Nonaka, I. and Takeuchi, H. (1995) *The Knowledge-Creating Company: How Japanese Companies Create the Dynamics of Innovation*, Oxford University Press, New York.
- Tackney, C.T., Sato, T. and Strömngren, O. (2010) 'Benchmarks in tacit knowledge skills instruction: the European Union-Research Oriented Participatory Education (EU-ROPE) model of Copenhagen Business School', *International Journal of Management in Education*, Vol. 4, No. 4, pp.463–485.
- Vedantam, S. and Stein, R. (2003) *Death Rate for Global Outbreak Rising*, 25 April, Washington Post, AO1.

Notes

- 1 For the double-blind peer review process, references to the school were generic: Scandinavian Business School. For journal publication and interests of transparency, we have specified the educational institution: Copenhagen Business School.
- 2 The Survey Monkey website is <http://www.surveymonkey.net>. Scanned images of the survey are given in Appendix 2.
- 3 See Morelli (2003) for an account of critical realism developments in the late 1920s when Lonergan was a graduate student in the UK and Morelli's careful exploration of possible influences from Sellars, Hicks, and others on Lonergan's later critical realism.
- 4 This practical guide, developed by one of the co-authors, is available on request.

Appendix 1

Initial reading and questions

Death rate for global outbreak rising

By Shankar Vedantam and Rob Stein

Washington Post Staff Writers

Friday, April 25, 2003; Page A01

The death rate for the worldwide outbreak of severe acute respiratory syndrome, which has fluctuated for months, has recently begun what looks like an ominous rise.

As of yesterday, 263 people have died of SARS worldwide. Although the rate of deaths from the strange new illness initially hovered as low as 3%, it has soared to twice as much in recent days, prompting questions about whether the virus is growing more lethal, spreading faster or striking more vulnerable people.

The questions are easier to ask than answer, because the epidemic is still in its early stages: The virus could indeed be killing more people than it did a few weeks ago, researchers say, possibly because of some unknown mutation that has made it more deadly. Equally likely, however, the virus's lethality may not be changing at all – and the changing numbers may simply reflect statistical aberrations or even changed definitions of what constitutes a case of the disease.

The death rate “has gone from 3.5 and 4 percent, and is now between 5.9 and 6 percent”, said Julie Gerberding, Director of the US Centers for Disease Control and Prevention, yesterday. “The fact that it is changing is not at all surprising, given that the case definition is changing.... Different countries are using different definitions”.

Still, the rising numbers are cause for concern for three reasons. First, the current – higher – death rate is statistically more reliable than the previous – lower – estimates.

Second, as hospitals learn to cope with the outbreak and doctors find ways to treat or stabilise patients, the death rate ought to head down, not up.

Finally, large numbers of cases so far, especially in places such as Hong Kong and Singapore, have involved hospital workers, who tend to be younger and healthier. As the SARS virus has spread to the general population in some places, it may strike more vulnerable elderly people and increase in lethality.

Statistics in Hong Kong show that the death rate among people younger than 55 is 3.6%. For patients between 65 and 75, it is 18.9%. For those older than 75, the death rate is 28.6%.

In Canada, the death rate yesterday as reported by the World Health Organization was 10.71%, one of the highest in the world, with most of the cases in the Toronto area. The number was based on WHO's count of 15 deaths and 140 cases.

"In Canada, the death rate seems to be very high", said David Brandling-Bennett, Deputy Director of the Pan American Health Organization in Washington, WHO's regional office". One would not expect to see higher death rates in Canada – it may depend on the intensity of the exposure. The more viral particles you get, the more severe the disease".

However, Andrew Simor, who has been treating SARS cases in Toronto, argued that the death rate in Canada is no higher than elsewhere. He said that communication lags meant that the WHO statistics are not always based on the latest figures: "We have 16 deaths out of 270 cases", he said, for a death rate of about 5.93%.

Still, Simor acknowledged, "We do have a sicker population", in part because a larger number of elderly people have been exposed. "Fifteen percent end up on a ventilator.... The older you are, the more likely you are to run into serious disease".

Such confusion and uncertainty is common in the early days of a worldwide epidemic such as SARS. Doctors do not yet have a practical laboratory test for the virus, meaning that some people with garden-variety influenza are probably being counted as SARS patients, while others, who really have the illness, are being left out of the calculations, said Marc Lipsitch, an epidemiologist at Harvard's School of Public Health.

Additionally, countries such as China, with the largest number of cases in the world, started out trying to mislead WHO about the true extent of SARS, placing national prestige and economic concerns over public health.

Finally, epidemiologists say that the overall number of SARS cases in the world has only recently soared to the size necessary to draw meaningful statistical conclusions about the death rate, which is calculated by dividing the number of deaths by the number of people infected.

Yesterday's statistics, for example, counted 4,439 cases of SARS and 263 deaths worldwide, for a death rate of 5.92%.

Lipsitch said a larger number of cases yields a more reliable rate because it reduces the possibility of chance variations. With 40 SARS deaths among any 1,000 cases, the calculated death rate would be 4%, but statistically, it is equally likely that in the next 1,000 cases, the number of deaths will be anywhere from 29 to 54 – yielding rates of 2.9% to 5.4%, he said. By contrast, with 4 deaths in just 100 cases, the rate for the next 100 cases is statistically likely to be anywhere from 1.1% to 9.9%.

Only two places, mainland China and Hong Kong, have enough cases to calculate meaningful local death rates: China's stands at 4.54%, while Hong Kong is at 7.32%.

One thing that worries health officials about SARS is that so far, it seems to be considerably more deadly than influenza, another major respiratory disease that has long afflicted humanity. SARS is the first life-threatening new disease to emerge in decades that can be spread from one person to another by casual contact.

Reliable death rates are available for the flu. In a typical year, it is usually below 1%. The Spanish flu pandemic of 1918 to 1919 had a death rate estimated to have been 2%. It, however, killed perhaps 25 million people worldwide because it quickly swept the globe and infected millions.

"What made it such a big killer was that so many people were infected", said Stephen S. Morse, Director of the Center for Public Health Preparedness at the Mailman School of

Public Health of Columbia University. “When you have such a large numerator, then you can still have a tremendous impact”.

The actual mortality rate from SARS is likely to be higher or lower than the 6% rate that is currently estimated. It may turn out that there are a large number of people who become infected but never fall ill. If that is the case, then the mortality rate will be much lower.

“We don’t know what the true denominator is. We don’t know the true number of actual SARS cases. So I think it’s a safe assumption that it’s going to change”, Morse said.

The transmissibility of a disease is another key factor in determining how much of a threat it poses. Some diseases have high mortality rates but are not easily spread, so the overall number of people who die is low. That is the case with Ebola, for example, which has a mortality rate of 70% to 95% but does not spread easily.

© 2003 The Washington Post Company

Questions for the SARS article

Please prepare these questions for our first session, if possible.

- 1 Why would a death rate rise? (Paragraph 1)
 - changes in the nature of the disease: increasingly virulent viral evolution/aberration
 - better estimators of the actual underlying illness, apart from any changes in the underlying illness
 - all of the above.
- 2 What are the ‘statistical aberrations’ referred to in the third paragraph? Be specific.
 - a simple outcome that is part and parcel of how the sampling distribution of the mean concept actually operates in practice.
- 3 Why is the current, higher death rate “statistically more reliable than the previous....?” (Paragraph 5) What does ‘reliable’ mean, in this instance?
 - reliability is validity over time
 - statistically more reliable measures are a function of sampling rigor: randomised and representative data selection steps.
- 4 Why is this statement true: “Lipsitch said a larger number of cases yields a more reliable rate because it reduces the possibility of chance variations?” (Paragraph 17) Alternatively, is the statement false? How can you be sure?
 - due our presumptive ‘faith’ in the sampling distribution of the mean and the Central Limit Theorem.
- 5 In paragraph 17, there are interesting estimates of possible death rates given sample size. How can they guess possible death rate variance (i.e., the possible range of death rates) in future samples if they know the sample size? Or is this just journalistic guesswork and utter non-sense?

- Because given the fundamental notions that enable statistical inference, one can 'know' the potential sample size needed for statistical testing at specific levels of confidence.

Appendix 2

Lesson plan and learning objectives

PhD course: applied quantitative methods for non-quantitative doctoral researchers in organisation and management studies

Schedule

Location:

CBS address, map, classroom and building location (Map previously sent via e-mail, for guest students).

Time plan:

Monday 10 May:

10.45–11.15 Registration, coffee and tea

11.15–13.15 Session 1: an overview of the place of quantitative studies in general empirical method:

- History, culture, and science
- Classical and statistical heuristic structures
- The complementarity of heuristic structures and insight

13.15–14.15 Lunch

14.15–15.45 Session 2: Descriptive statistics and introduction to SPSS (hands on)

15.45–16.00 Break (coffee and tea are provided)

16.00–17.00 Session 2: Introduction to SPSS (hands on) – continued

Tuesday 11 May:

9.00–10.30 Session 3: A session of review and/or remediation: statistics as description and for inference, how these differ: descriptive stats and t-test

10.30–10.45 Break (coffee and tea are provided)

10.45–12.15 Session 4: Explanation of statistical models (correlation, regression analyses: OLS and logit)

12.15–13.15 Lunch

13.15–15.00 Session 5: Estimation and interpretation of statistical models (t-test, correlation, OLS and logit) (hands on)

15.00–15.15 Break (coffee and tea are provided)

15.30–17.15 Session 6: Class discussions and/or individual sessions on the application of quantitative methods to individual research questions (brief presentation of individual projects and discussion)

Wednesday 12 May:

- 9.00–10.30 Session 7: Estimation and interpretation of statistical models (correlation, OLS and logit) (hands on) – continued
- 10.30–10.45 Break (coffee and tea are provided)
- 10.45–12.15 Session 8: Explanation, estimation and interpretation of statistical models (factor analysis)
- 12.15–13.15 Lunch
- 13.15–15.30 Session 9: Part 1 or the usefulness of factor analysis – a talk by *a guest Lecturer, Associate Professor, CBS*.
Part 2 or evaluation of the course and class discussions on the application of quantitative methods to individual research questions and debriefing

Objectives:

At the end of the course, doctoral students should be able to:

- a know and understand the historical and cultural contexts within which contemporary research methods function
- b specify the complementarities of qualitative and quantitative research within the general empirical method
- c know the quantitative approaches appropriate to their specific research interests
- d use statistical packages needed for their doctoral research needs
- e evidence a nuanced ability to consider empirical research questions in organisational and management studies, so they may
- f better understand empirical literature and read it critically, in order to
- g suggest appropriate quantitative methods to address research questions.

Appendix 3

The pre-course survey measure

1. Introduction

This survey will take about 10 minutes and will provide useful information for the course. It will aid us to focus on particular research interests and issues of a quantitative nature that concern your doctoral studies.

1. What was your undergraduate major?

2. Do you have a Master's degree?

☐ Yes.

☐ No.

What was the Master's in?

3. How many semesters of statistics have you had in your academic career to date?

4. Is the difference between descriptive and inferential statistics clear to you?

☐ Yes.

☐ No.

2. To your specific research interests

1. What is the research question you would like to answer by applying quantitative analysis?

2. If you already have a research method or design in mind, what is it?

Questions about the data you propose to analyze:

3. Primary or secondary data?

- ☐ Primary.
- ☐ Secondary.
- ☐ Both.
- ☐ I honestly don't know the difference.

4. If you have a data set, does it have a name, such as a specific survey?

- ☐ Yes.
- ☐ No.
- ☐ I can't tell you.

If yes, what is it?

5. Do you already have access to this data?

- ☐ Yes.
- ☐ No.

Comment?

6. What is your proposed dependent variable (the variable of interest)?

Please provide the phrasing for this based on the question you may have used in any questionnaire, along with possible answers.

7. How would you characterize this dependent variable?

- ☐ Nominal.
- ☐ Ordinal.
- ☐ Metric.
- ☐ I'm clueless.
- ☐ That's what you are going to help me do!

8. Can you list the possible independent variables you think account for the observed variance in your dependent variable?

If you can, a short description of them and their possible scaling would be interesting.

3. Concluding questions and additional comment space

1. Do you wish to have an extensive set of background readings on research method, particularly quantitative methods, or do you wish to simply focus upon your particular research concerns?

- ☐ An extensive research readings list would help.
- ☐ I just wish to focus on my doctoral research issue.
- ☐ Why both, of course!

2. May we share this information with the class?

- ☐ No, this is strictly for the instructors and should not be shared.
- ☐ Yes, go right ahead and share this.

3. Any further comment or request regarding the course?

4. What is your name and email address?

That's it! Thank you very much and we'll see you soon.

Appendix 4

Annotated bibliography on the critical realism of Bernard J.F. Lonergan

Bernard J.F. Lonergan, S.J., was a polymath whose published scholarship on philosophy, theology, economics, and education extends to 25 collected works published by the University of Toronto Press. Here is a modest and brief annotated bibliography of course texts and additional sources.

1 Course texts

- Lonergan, Bernard, J.F. S.J. (1992) *Insight: A Study of Human Understanding*, 5th ed., edited by F.E. Crowe, S.J. and R.M. Doran, S.J., Vol. 3 in the Collected Works of Bernard Lonergan, University of Toronto, Toronto.

Lonergan's major work in epistemology. We cite brief sections in course material. Interested readers who have backgrounds in math, physics, and the natural sciences may proceed from start to finish. Qualitative researchers may find it easier to read the Preface and Introduction, resume the material on statistics, portions of which are presented in this course, and then proceed directly to Chapter 6, 'Common Sense and its Subject'.

- Lonergan, Bernard, J.F. S.J. (1971) *Method in Theology*, 2nd ed., University of Toronto Press, Toronto.

This work extends insight into the field of theology. While seemingly remote from the social sciences, the text is very clearly written and offers a useful context to

reflect on the nature and role of the social sciences, whether or not one believes in God.

- Lonergan, Bernard, J.F. S.J. (1996) 'Theology in its new context', in Ryan, W.F.J. S.J., and Tyrrell, B.J. S.J. (Eds.): *A Second Collection: Papers by Bernard J.F. Lonergan, S.J.*, Chapter 5, pp.55–68, University of Toronto, Toronto.

This brief, readable chapter forcefully outlines the significance of science from the larger perspective of Western culture, to its implications for theology.

- Morelli, M.D. (1999) 'Lonergan's unified theory of consciousness', *Method*, Vol. 17, pp.171–188.

Presents the epistemological significance of Lonergan's endeavour in reference to the fact of consciousness. And then references this to neuroscientific studies, associationist approaches (Hobbes, Hume, and Locke), and phenomenology (Brentano and Husserl).

2 Additional readings and resources

A Texts

- Lonergan, Bernard, J.F. S.J. (1996) 'Insight revisited', in Ryan, W.F.J. and Tyrrell, B.J. (Eds.): *A Second Collection: Papers by Bernard J.F. Lonergan, S.J.*, pp.263–278, University of Toronto, Toronto.

A brief, reflective resume of Lonergan's major work in epistemology, by the very author. An excellent piece to read before embarking on the larger text.

- Lonergan, Bernard, J.F. S.J. (1988) 'Cognitive structure', in Crowe, F.E. and Doran, R.M. (Eds.): *Collection*, Vol. 4, pp.205–211, in the Collected Works of Bernard Lonergan, University of Toronto Press, Toronto.

One short chapter on one key concept for students of the social sciences.

- Morelli, M.D. and Morelli, E.A. (1997) *The Lonergan Reader*, University of Toronto Press, Toronto.

A focused compilation of Lonergan's works.

- Crowe, F.E. S.J. and Doran, R.M. S.J. (Eds.) *The Collected Works of Bernard Lonergan. The Robert Mollot Collection*, University of Toronto, Toronto, available at http://www.lonergan.concordia.ca/bib/collected_works.htm, also available through Amazon.com.

B Websites

- 1 Bernard Lonergan archive: resources in Lonergan studies, Marquette University, WI, USA, available at <http://www.bernardlonergan.com/>.

An archival site for primary sources, including scans of the author's original drafts.

- 2 Lonergan resource, Marquette University, WI, USA, available at <http://www.lonerganresource.com/>.

A site for secondary sources on Lonergan studies.